A. Doremy's Paint

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: greedy | https://codeforces.com/problemset/problem/1764/A

Doremy has n buckets of paint which is represented by an array a of length n. Bucket i contains paint with color a_i.

Let $c(l,r)$ be the number of distinct elements in the subarray $[a_l,a_{l+1},\ldots,a_r]$. Choose 2 integers l and r such that $l \le r$ and $r-l-c(l,r)$ is maximized.

Input

The input consists of multiple test cases. The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. The description of the test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 10^5$) — the length of the array a.

The second line of each test case contains n integers $a_1,a_2,\ldots,a_n$ ($1 \le a_i \le n$).

It is guaranteed that the sum of n does not exceed 10^5.

Output

For each test case, output l and r such that $l \le r$ and $r-l-c(l,r)$ is maximized.

If there are multiple solutions, you may output any.

Example

Input

```
7
5
1 3 2 2 4
5
1 2 3 4 5
4
2 1 2 1
3
2 3 3
2
2 2
1
1
9
9 8 5 2 1 1 2 3 3
```

Output

```
2 4
1 5
1 4
2 3
1 2
1 1
3 9
```

Note

In the first test case, a=[1,3,2,2,4].

- When l=1 and r=3, $c(l,r)$=3 (there are 3 distinct elements in [1,3,2]).
- When l=2 and r=4, $c(l,r)$=2 (there are 2 distinct elements in [3,2,2]).

It can be shown that choosing l=2 and r=4 maximizes the value of $r-l-c(l,r)$ at 0.

For the second test case, a=[1,2,3,4,5].

- When l=1 and r=5, $c(l,r)$=5 (there are 5 distinct elements in [1,2,3,4,5]).
- When l=3 and r=3, $c(l,r)$=1 (there is 1 distinct element in [3]).

It can be shown that choosing l=1 and r=5 maximizes the value of $r-l-c(l,r)$ at -1. Choosing l=3 and r=3 is also acceptable.